Day 3: DNS Server ConfigurationTheory

Task 1: Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

Requirements:

- Ubuntu Linux
- BIND9 DNS Server
- Terminal access

Commands Used:

Update package information:

sudo apt update

Install BIND9 and DNS utilities:

sudo apt install bind9 bind9-utils dnsutils -y

Check DNS service status:

sudo systemctl status bind9

DNS Zone Configuration:

Open the configuration file:

sudo nano /etc/bind/named.conf.local

Add the following zone configuration:

```
zone "musicmania.in" {
   type master;
   file "/etc/bind/db.musicmania.in";
};
```

This configuration defines musicmania.in as a primary DNS zone managed by the local BIND9 server.

Result: A DNS server is installed and configured to manage the musicmania.in domain.

Task 2: Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.

To create an A record that maps www.musicmania.in to the IP address 192.168.1.50.

DNS Zone File Configuration:

Create the zone file:

sudo nano /etc/bind/db.musicmania.in

Add the following records:

```
$TTL 86400
@  IN  SOA ns1.musicmania.in. admin.musicmania.in. (
     2026110901
     3600
     1800
     604800
     86400
)

@     IN  NS     ns1.musicmania.in.
ns1    IN  A      127.0.0.1
www    IN  A      192.168.1.50
```

Explanation:

An A record maps a domain name to an IPv4 address. In this configuration, www.musicmania.in is associated with 192.168.1.50.

Verification Command:

nslookup www.musicmania.in 127.0.0.1

Expected Result:

The DNS server should return the IP address 192.168.1.50 for www.musicmania.in.

Task 3: Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

To create a CNAME record for play.musicmania.in pointing to www.musicmania.in.

Configuration:

Add the following record to the DNS zone file:

play IN CNAME www.musicmania.in.

The CNAME record creates an alias for an existing domain name.

Verification Command:

nslookup play.musicmania.in 127.0.0.1

Expected Result:

The DNS server should resolve play.musicmania.in through its alias www.musicmania.in and return the IP address 192.168.1.50.

Explanation:

CNAME records allow multiple domain names to point to the same canonical hostname. This avoids the need to maintain separate A records for every alias.

Task 4: Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.

To identify and troubleshoot problems when the DNS server fails to resolve musicmania.in.

Step-by-Step Troubleshooting:

1. Check DNS Configuration Files

Open the main zone configuration file:

sudo nano /etc/bind/named.conf.local

Verify that the zone name and zone file path are correct.

2. Check Zone File Syntax

Run:

sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in

This command checks the syntax and validity of the DNS zone file.

3. Check Main BIND Configuration

Run:

sudo named-checkconf

This command identifies syntax errors in BIND configuration files.

4. Restart the DNS Service

Run:

sudo systemctl restart bind9

Restarting the service applies configuration changes.

5. Check DNS Service Status

Run:

sudo systemctl status bind9

Verify that the service is active and running.

6. Test Domain Resolution

Run:

nslookup www.musicmania.in 127.0.0.1

Alternatively:

dig @127.0.0.1 www.musicmania.in

These commands help verify whether the DNS server is returning the correct IP address.

7. Check DNS Logs

Run:

sudo journalctl -u bind9 -n 50

This command displays recent DNS service logs and helps identify configuration or service-related errors.

Common DNS Problems and Solutions:

Problem	**Possible Solution**
Incorrect zone configuration	Check named.conf.local
Zone file syntax error	Use named-checkzone
DNS service stopped	Restart BIND9
Incorrect IP address	Verify A record
DNS query failure	Test using nslookup or dig